



FIRST SEMESTER 2024-2025
COURSE HANDOUT

Date: 30.07.2024

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No. : CE F230
Course Title : Civil Engineering Materials
Instructor-in-charge : Dr. Mukund Lahoti

1. Objective of the course and Course Description:

The course deals with properties, characterization, testing, selection and quality control of construction material. For economical, durable and high performance of structure, it is imperative to have knowledge on construction materials and its wider applications. This course provides the basic and enhanced overview on various construction materials presently used in practice.

2. Text Books:

- T1. Gambhir, M.L. (2013) “Concrete Technology” Tata McGraw-Hill Publishing Company Ltd, 5th edition.
T2. Duggal, S.K. (2012) “Building Materials” New Age International Pvt. Ltd., 4th edition.
T3. Gambhir, ML and Jamwal, N. (2014) “Lab Manual- Building and Construction Materials (Testing and quality control)”, Mc Graw Hill, 1st edition (**Lab Manual**)

3. Reference Books:

- R1. Relevant B.I.S, IRC, ASTM codes
R2. Mehta, P.K. and P.J.M. Monteiro. (2006) “Concrete Microstructure, Properties, and Materials”, The McGraw-Hill Companies, United States.
R3. Neville, A.M. (1995) “Properties of Concrete”, Pearson Education, 4th edition
R4. Gambhir, M.L., and Jamwal, N. (2011) “Building Materials- Products, Properties and Systems”, Mc Graw Hill, 1st edition
R5. Moondra, HS and Gupta, R. (2000) “Lab Manual for Civil Engineering”, CBS, 2nd edition (**Lab Manual**)

4. Course Plan:

Module No.	Lecture Session	Reference	Learning outcome
1	Civil engineering materials- An overview	Slides	Understand various construction materials, the importance of concrete, process of cement manufacturing, cement hydration, microstructures, and SCMs
	Concrete as a construction material	1, T1	



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	Cement and hydration, water/cement ratio and supplementary cementitious materials	2, T1 and 5, T2	
2	Aggregate classification and tests	3, T1 and 6, T2	Understand important aggregates sources, their classification, influence of properties of aggregates on their performance, testing methods/standards
3	Chemical admixtures	5, T1	Develop clear understanding of the purpose and mechanism of chemical admixtures
4	Concrete mix design	IS10262, 10, T1	Design concrete mix and explain the effect of concrete composition on concrete fresh properties, transportation and handling of concrete
	Fresh Concrete	6, 7 T1	
	Working with fresh concrete	11, 12, T1	
5	Hardened concrete properties	8, T1	Explain the effect of composition on hardened concrete properties and various concrete tests including NDTs
	Testing on hardened concrete	8, 13 T1	
6	Durability of concrete	15, T1 and slides	Examine various durability issues in concrete and recommend preventive measures
7	Special Concretes	14, 16, T1	Categorize special concretes and understand features of their mix design
8	Clay products: Properties, tests, selection	2, T2	Describe important ingredients, manufacturing and classification of bricks. Define important terms in brick masonry. Experiment key properties of bricks.
9	Wood and timber	4, T2	Indicate salient parts of timber in X-section of wood. List defects. Applications.
10	Steel, ferrous and non-ferrous metals	14, 15, T2	Describe manufacturing of steel, classification, applications. Prepare comprehensive list of important sources, properties and usage of aluminum, copper, zinc, lead, tin and nickel.



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11	Lime: Properties, tests, application	8, T1	List steps involved in lime manufacture, impurities & effects, lime properties, tests and applications.
12	Introduction to highway materials	19, T2, Relevant IS and IRC codes	Understand concisely various materials involved in highway construction
13	Polymeric material, Geo-synthetics and Construction equipments	21, T2, Slides	Understand important properties and applications of FRPs, FRCs, and geosynthetics. Glimpse to significant construction equipments.

5. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-semester exam	90 min	25	3 rd Oct, FN 1	CB
Tutorials	-	15	-	CB
Laboratory	-	20	Lab record (10%) + Lab quiz (10%) date will be announced	OB + CB
Comprehensive Examination	180 min	40	2 nd Dec, FN	CB + OB

7. Consultation Hour: Tuesday 4-5 pm. Please email at mukund.lahoti@pilani.bits-pilani.ac.in if needed.

8. Notices: Announcements and course material will be communicated via Nalanda.

9. Make-up Policy:

- Take prior permission from Instructor in charge.
- Make-up will be granted on a case by case basis only on genuine reasons.
- For medical cases, a certificate from the concerned physician of the Medical Centre must be produced.
- No make-up for tutorials.
- One buffer for lab.

10. Note (if any): If you miss a complete component (Lab/Tutorial), you may be granted NC.

Instructor-in-charge
Course No. CE F230



Civil Engineering Materials (Laboratory)

Instructor-in-charge : Dr. Mukund Lahoti

Text Books:

1. Gambhir, ML and Neha Jamwal, Lab Manual- Building and Construction Materials (Testing and quality control), Mc Graw Hill, 1st edition 2014
2. Moondra, HS and Rajiv Gupta, Lab Manual for Civil Engineering, CBS 2nd edition 2000

LIST OF EXPERIMENTS

Week	Name of experiment
1	Introduction to all experiments, marking scheme, lab report format, etc.
2	Brick: (a) Compressive strength, efflorescence tests (b) Making fly ash bricks
3	Cement: (a) Heat of hydration of cement (b) Test brick samples for compressive strength and efflorescence
4	Cement: (a) Normal consistency and setting time (b) Fineness of cement
5	Fine aggregate: (a) Bulking of sand (b) Silt content (c) Specific gravity
6	Fineness modulus and grain size distribution: Fineness modulus and grain size distribution
7	Cement mortar: ○ Cast 3 cubes for compressive strength testing (air dried in shade, water-cured) ○ Cast 3 briquettes for tensile strength testing
8	Concrete (casting): (a) Nominal mix design ○ Cast 2 cylinders, 2 cubes, 1 beam ○ Test workability (slump)
9	Concrete (casting): IS 10262 mix design ○ Cast 2 cylinders, 2 cubes, 1 beam ○ Test workability (slump)
10	Concrete testing: (a) Modulus of rupture (beam test) (b) Test compressive strength of cube and cylinder (c) Split tensile strength of cylinder
11	Demo to various tests on: Rapid Chlorine Penetration Test (RCPT), Non-Destructive Testing (NDT), tensile test on steel